

Disproof of TLMC Conjecture 00000000477

Promotion orbit lengths on linear extensions

1 Verdict

FALSE. The lcm of promotion orbit lengths need not divide $\#LE(P)$.

Conjecture (00000000477). *Let P be a finite poset and let promotion act on the set $LE(P)$ of its linear extensions. Then the lcm of the promotion orbit lengths divides $\#LE(P)$.*

2 The counterexample

Take P to be the Ferrers poset of the Young diagram of shape $(3, 2)$: its five elements are the cells $(0, 0), (0, 1), (0, 2), (1, 0), (1, 1)$ of the diagram, ordered componentwise, i.e. $(i, j) \leq (i', j')$ iff $i \leq i'$ and $j \leq j'$.

Linear extensions of P are in bijection with standard Young tableaux (SYT) of shape $(3, 2)$: the entry of a cell is its position in the extension. The hook length formula gives

$$\#LE(P) = \frac{5!}{4 \cdot 3 \cdot 1 \cdot 2 \cdot 1} = 5,$$

and indeed exactly five tableaux exist:

t_1	t_2	t_3	t_4	t_5
1 2 3	1 2 4	1 2 5	1 3 4	1 3 5
4 5	3 5	3 4	2 5	2 4

3 Promotion and its orbits

Apply Schützenberger promotion (remove the corner entry, jeu-de-taquin slide the hole to an outer corner, write 6 in the hole, decrease all entries by 1). Working out the jeu de taquin for this fixed shape (the hole makes at most two slides), one obtains

$$t_1 \mapsto t_3 \mapsto t_4 \mapsto t_1, \quad t_2 \mapsto t_5 \mapsto t_2.$$

For instance, promoting $t_1 = (1, 2, 3, 4, 5)$: the right neighbour 2 of the hole is smaller than the lower neighbour 4, so 2 slides up; then of 3 (right) and 5 (below), the smaller 3 slides; the hole reaches the outer corner $(0, 2)$, receives 6, and decreasing by 1 yields $(1, 2, 5, 3, 4) = t_3$.

So promotion partitions the five linear extensions into two orbits, of lengths 3 and 2:

$$\{t_1, t_3, t_4\}, \quad \{t_2, t_5\}.$$

4 The falsification

$$\text{lcm}(3, 2) = 6, \quad 6 \nmid 5 = \#LE(P).$$

Hence the conjecture fails already for this five-element poset.

5 Verification

- `reproduce.py`: an independent brute-force recomputation. It enumerates all $5!$ orderings of the cells, keeps the linear extensions (5 of them), converts them to tableaux, applies Schützenberger promotion, and recovers the orbit lengths $[2, 3]$, lcm 6, and $6 \nmid 5$. Run with `python3 reproduce.py`; assertions check every number.
- `lean4/`: a core-Lean-4 (no Mathlib) certificate. `Main.lean` enumerates the 5^5 candidate tuples in the kernel and proves `syts = [t1,...,t5]` and `syts.length = 5`; promotion is implemented explicitly and the orbit equations $t_1 \rightarrow t_3 \rightarrow t_4 \rightarrow t_1$, $t_2 \rightarrow t_5 \rightarrow t_2$ plus $\text{lcm}(3, 2) = 6$ and $6 \nmid 5$ are proved by `decide`. `Check.lean` audits every theorem with `#print axioms`: all report *does not depend on any axioms* (no `propext`, no `Quot.sound`, no `Classical.choice`), no `sorry`.

6 Scope and boundary

The conjecture is stated for arbitrary finite posets, so the single counterexample above refutes it. It is worth recording where the divisibility does hold: for rectangular shapes (a^b) promotion is known (Reiner–Stanton–White cyclic sieving; Haiman) to have a single orbit of length ab , whose lcm ab trivially divides the number of SYT only in special cases — the point here is merely that the non-rectangular shape $(3, 2)$ breaks the claimed divisibility, with orbit lengths $3 + 2 = 5 = \#LE(P)$ whose lcm 6 exceeds $\#LE(P)$ itself.